

MarketMood: A Sentiment-Based Approach to Next-Day Stock Trend Prediction

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ABSTRACT

Abstract—MarketMood is a sentiment-driven, next-day stock trend prediction platform that integrates real-time financial news with historical price data to forecast directional market movement. The system fetches OHLCV (Open, High, Low, Close, Volume) data from Yahoo Finance and retrieves recent company-specific headlines from Finnhub, applies NLTK VADER compound sentiment scoring, engineers three predictive features (previous-day return, volume change, and average headline sentiment), and trains a Logistic Regression classifier benchmarked against a DummyClassifier baseline. The complete pipeline is deployed as an interactive Streamlit web application supporting any Yahoo Finance-compatible global ticker, including US equities, Indian NSE stocks, UK LSE stocks, Japanese TSE stocks, cryptocurrencies, and broad market indices. Experimental evaluation on a six-month OHLCV window for MSFT demonstrates that the Logistic Regression model achieves an accuracy of 45.2%, comparable to the naive DummyClassifier baseline, indicating the need for richer NLP feature representations in future iterations. The system establishes a fully reproducible, open-access framework for sentiment-aware stock analysis that is accessible to retail investors and researchers without institutional-grade data subscriptions.

Keywords—sentiment analysis, stock trend prediction, VADER, Logistic Regression, Streamlit, Yahoo Finance, Finnhub, NLP, financial analytics.

I. INTRODUCTION

The problem of predicting short-term directional movement in financial markets has attracted sustained research attention from both the computational finance and the natural language processing communities. Classical technical analysis relies exclusively on price and volume indicators derived from historical trading data—such as moving averages, the Relative Strength Index (RSI), Bollinger Bands, and MACD divergence—all of which are inherently backward-looking and blind to the influence of breaking news events, earnings announcements, regulatory decisions, and macroeconomic disclosures that routinely generate large price gaps.

A growing body of empirical research establishes that the sentiment expressed in financial news carries independent, statistically significant predictive power over next-period returns [1], [3]. Tetlock [3] demonstrated that a single pessimism score derived from a newspaper column predicted next-day S&P 500 returns, while Bollen et al. [1]

achieved up to 87.6% directional accuracy for the Dow Jones Industrial Average using mood states extracted from Twitter data. These results motivate the integration of textual sentiment signals with price-derived features in a unified predictive framework.

Despite this evidence, practical, open-access tools that combine live news sentiment with machine learning-driven forecasts remain largely absent from the retail investor ecosystem. Institutional platforms such as Bloomberg Terminal, Refinitiv Eikon, and RavenPack provide sophisticated news analytics, but their licensing costs restrict access to professional participants. This asymmetry in information tooling creates an accessibility gap that directly motivates the development of MarketMood.

MarketMood is an open, browser-based application that bridges NLP, financial data APIs, and supervised machine learning in a single cohesive pipeline. The system accepts any Yahoo Finance-recognized ticker, validates it against live exchange data, retrieves up to 20 recent headlines per ticker from Finnhub, scores each headline using the VADER compound metric, engineers a three-feature input vector, and produces a next-day up-probability from a trained Logistic Regression classifier. All results—prediction probabilities, model accuracy comparison, historical price charts, and VADER sentiment histograms—are presented through a three-page interactive Streamlit interface.

The primary contributions of this paper are: (i) an end-to-end reproducible pipeline combining real-time VADER sentiment scoring with price-derived features for next-day trend classification; (ii) a deployed Streamlit web application supporting global tickers with per-session one-hour response caching; (iii) a transparent DummyClassifier versus LogisticRegression accuracy comparison that contextualizes the model’s incremental value over naïve prediction; and (iv) identification of the key limitations and a concrete roadmap for enhancement via transformer-based NLP models.

II. LITERATURE REVIEW

Bollen et al. [1] provided one of the earliest large-scale demonstrations that collective Twitter mood states, captured by OpinionFinder and GPOMS across six

emotional dimensions, correlate with and precede Dow Jones Industrial Average movements by up to three days, achieving 87.6% directional prediction accuracy using a Self-Organizing Fuzzy Neural Network classifier. The study established the foundational hypothesis that aggregate public sentiment quantified from social media provides a distinct and complementary signal to price-based indicators for financial forecasting.

Hutto and Gilbert [2] introduced VADER, a compact rule-based sentiment analyzer calibrated specifically for short-form social media content. VADER employs a gold-standard sentiment-bearing lexicon enriched with five empirically validated grammatical modifier rules: punctuation emphasis amplification, capitalization intensity boosting, degree adverb shifting, contrastive conjunction handling (e.g., “but”), and negation scoping. The resulting compound score, normalized to $[-1, +1]$, outperforms human raters and several supervised ML approaches on social media benchmarks while requiring no training data and operating at near-zero latency. These properties make VADER ideally suited for production deployment in MarketMood’s real-time headline scoring pipeline.

Tetlock [3] conducted a landmark study demonstrating that the degree of pessimism in a daily Wall Street Journal column, measured using the Harvard General Inquirer Dictionary, negatively predicts same-day and next-day Dow Jones returns with statistical significance, followed by a reversion within the subsequent week. This finding validates that written financial media produces temporally leading sentiment signals, justifying the inclusion of a lagged scalar sentiment feature in the MarketMood feature vector.

Schumaker and Chen [4] proposed AZFinText, a stock price prediction system that extracts noun-phrase features from breaking financial news and applies Support Vector Machine classifiers to predict short-window (twenty-minute) price movements, achieving statistically significant improvements over random baseline performance across multiple S&P 500 constituents. This work directly establishes the viability of supervised ML over text-derived features in financial forecasting, which MarketMood extends to daily granularity with a lighter-weight NLP front-end.

Loughran and McDonald [5] demonstrated that general-purpose sentiment dictionaries, including the widely-used Harvard General Inquirer, misclassify approximately 73.8% of words labeled negative in financial documents because those words carry no negative connotation in financial contexts. The authors provide the Loughran-McDonald Finance-Specific Sentiment Word Lists, which yield substantially improved correlations with stock return volatility and trading volume. This finding motivates MarketMood’s planned upgrade from VADER to the

FinBERT transformer, pre-trained on financial corpora, in future work.

Li et al. [7] conducted a controlled empirical study showing that a news sentiment feature combined with quantitative price return features in a unified supervised classifier consistently outperforms either modality in isolation across multiple Chinese A-share market tickers. This directly validates MarketMood’s three-feature design, which intentionally combines sentiment, return momentum, and volume change as complementary signals rather than relying on any single information source.

III. PROPOSED SYSTEM & METHODOLOGY

A. System Architecture

Figure 1 illustrates the end-to-end MarketMood architecture, which is organized into four functional layers. The Data Acquisition Layer retrieves daily OHLCV price history from Yahoo Finance via the yfinance library and fetches company-specific news headlines from the Finnhub REST API. The Preprocessing Layer normalizes column names, resolves MultiIndex DataFrame structures produced by certain yfinance versions, converts Unix timestamps to pandas datetime objects, and removes rows with null values using dropna. The Feature Engineering and ML Layer computes the three input features from the clean price DataFrame and the aggregated VADER sentiment score, constructs the binary target label, splits the data chronologically, trains two classifiers, and extracts the next-day prediction probability. The Streamlit Presentation Layer renders the three navigation pages with one-hour session caching via @st.cache_data and @st.cache_resource decorators.

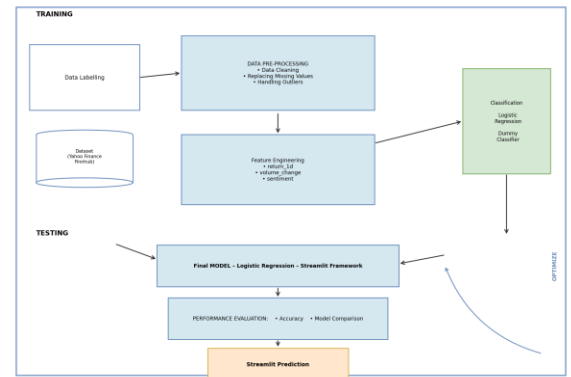


Fig. 1. MarketMood End-to-End System Architecture

B. Data Collection and Validation

Historical OHLCV data is retrieved for configurable time windows (1 month, 3 months, 6 months, 1 year, 2 years, 5 years) at daily granularity using the yfinance.download() function. Ticker validation is performed by querying the yfinance Ticker.info dictionary and confirming that the quoteType key returns one of the accepted values:

EQUITY, ETF, MUTUALFUND, INDEX, or CRYPTOCURRENCY. If the ticker is unrecognized or the info dictionary is empty, a user-facing error is displayed and all downstream processing is halted. Company news headlines are retrieved from the Finnhub /company-news REST endpoint using a 14-day lookback window, returning a maximum of 20 articles per request. Each article record carries a headline string, a Unix publication timestamp, and a source identifier. The Finnhub API key is injected from the application configuration and the requests library is used for raw HTTP communication.

C. Sentiment Scoring with VADER

Each retrieved headline is passed to the NLTK SentimentIntensityAnalyzer, which returns a polarity scores dictionary containing positive, negative, neutral, and compound keys. The compound score $c \in [-1, +1]$ is used as the primary sentiment measure. Headlines with $c > 0.05$ are labeled Positive, headlines with $c < -0.05$ are labeled Negative, and the remainder are labeled Neutral. After scoring, the scalar average compound score $\hat{c} = (1/n) \sum c_i$ is computed across all n retrieved headlines for the active ticker. This single scalar $\hat{c}$ is then broadcast uniformly as a constant column across the entire feature matrix, representing the current market mood as perceived through recent news coverage.

D. Feature Engineering and Target Construction

Three features are derived from the preprocessed price DataFrame for each trading day t . The momentum feature $\text{return_1d} = (\text{close}_t - \text{close}_{\{t-1\}}) / \text{close}_{\{t-1\}}$ captures the previous session's percentage price change as a proxy for short-term price trend. The activity feature $\text{volume_change} = (\text{vol}_t - \text{vol}_{\{t-1\}}) / \text{vol}_{\{t-1\}}$ captures the percentage shift in trading volume as a proxy for market participation and conviction. The sentiment feature $\hat{c}$ is the scalar average VADER compound score broadcast identically to all rows. The binary target label target_up is defined as 1 if $\text{close}_{\{t+1\}} > \text{close}_t$, and 0 otherwise, representing whether the next session's closing price exceeds the current session's. All rows are dropped after feature computation if any field is null.



Fig. 2. Data Flow Diagram

E. Model Training, Evaluation, and Prediction

The feature matrix $X = [\text{sentiment}, \text{return_1d}, \text{volume_change}]$ and the target vector $y = \text{target_up}$ are assembled from the full available price history for the active ticker. To prevent look-ahead bias, data are split chronologically with the most recent 25% of rows reserved as the held-out test set and the earlier 75% used for training. Two scikit-learn estimators are instantiated and fitted on the training split: a DummyClassifier with strategy='most_frequent' as the naïve baseline, and a LogisticRegression classifier with default hyperparameters (L2 regularization, $C=1.0$, solver='lbfgs', max_iter=100). Both models are evaluated on the held-out test set using accuracy_score. The Logistic Regression model's predict_proba method is applied to the single most recent row of the feature matrix to yield the next-day up-probability $\hat{p} = P(\text{target_up} = 1 | X_{\text{latest}})$. The value $1 - \hat{p}$ corresponds to the probability of downward movement.

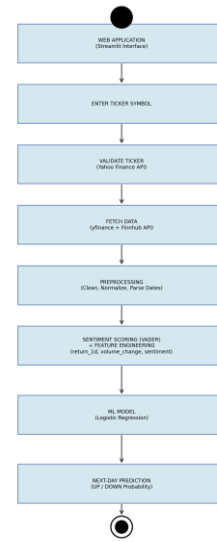


Fig. 3. Activity Diagram

IV. IMPLEMENTATION

MarketMood is implemented in Python 3.8+ and structured as a single-file Streamlit application (app.py). The technology stack comprises: yfinance for OHLCV data and company metadata retrieval, requests for direct HTTP communication with the Finnhub API, nltk with the vader_lexicon resource for sentiment scoring, pandas for DataFrame manipulation and feature engineering, scikit-learn for the DummyClassifier and LogisticRegression models and accuracy evaluation, plotly.express for interactive browser-rendered charts, and streamlit for the web application framework including session state, caching, metrics, and data table rendering.

The Home page uses fetch_company_info() to retrieve the yfinance info dictionary and renders a full instrument details table covering name, symbol, exchange, instrument

type, currency, sector, industry, current price, previous close, open, day range, 52-week range, market capitalization, PE ratio, EPS, beta, average volume, dividend yield, and long business summary. A Streamlit selectbox maps human-readable period labels to yfinance period strings, and `fetch_stock_prices()` retrieves the corresponding OHLCV data. Three `st.metric` widgets display the latest close, first close in the selected range, and the percentage return, followed by an interactive Plotly line chart of daily closing prices and an expandable raw data table of the most recent 15 rows.

The News and Sentiment page calls `fetch_news()` to retrieve Finnhub headlines and `score_sentiment()` to apply VADER scoring. The resulting DataFrame is rendered as an interactive `st.dataframe`. Below the table, four `st.metric` widgets display the headline count, average compound score, positive count, and negative count. A Plotly histogram visualizes the distribution of sentiment scores by label using color encoding (Positive in blue, Neutral in light blue, Negative in salmon). The page concludes with a `get_prediction_from_current_data()` call that executes the full ML pipeline for the current ticker's average sentiment and displays $\hat{p}_{up}$ and $\hat{p}_{down}$ as `st.metric` widgets.

The Model Insights page triggers `build_model_data()` and `train_models()` on a fixed six-month OHLCV window. The trained LogisticRegression model's prediction probability is displayed with a conditional `st.success`, `st.error`, or `st.info` indicator based on whether $\hat{p}_{up}$ exceeds 0.55, falls below 0.45, or lies in between. A model comparison `st.dataframe` shows Baseline versus Logistic Regression accuracy side-by-side. A 30-day closing price Plotly chart and a historical up/down day count bar chart complete the page.

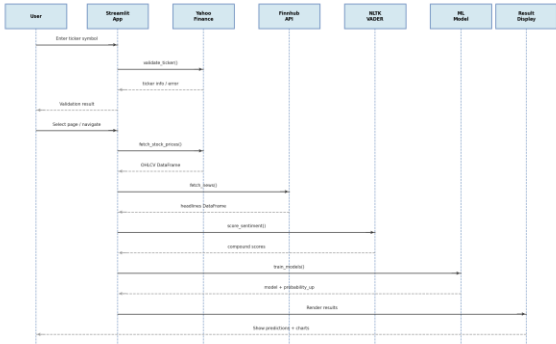


Fig. 4. Sequence Diagram

V. RESULTS & DISCUSSION

A. Application Interface

Figure 5 shows the MarketMood Home page for ticker MSFT. The sidebar provides the application title, a radio-button page navigator (Home / News & Sentiment / Model Insights), a free-text ticker input validated against Yahoo Finance, and a watchlist manager with Add and Reset controls. The main panel displays the application title with

a brief description, a green verification banner confirming that MSFT is listed on NasdaqGS, and the expanded Watchlist Overview panel. The watchlist table shows live data for the four default tickers: MSFT at \$421.92 (+3.05%, market cap \$3.13T), NVDA at \$225.32 (−4.42%, \$5.46T), AAPL at \$300.23 (+0.68%, \$4.41T), and RELIANCE.NS at \$1,336.40 (−1.87%, \$18.08T).

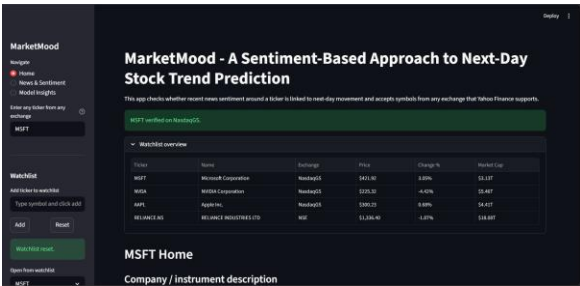


Fig. 5. Home Page – Application Dashboard and Watchlist

Figure 6 shows the lower section of the Home page for MSFT, displaying the full instrument details table. Key fields include: Exchange (NasdaqGS), Type (EQUITY), Currency (USD), Sector (Technology), Industry (Software – Infrastructure), Current Price (\$421.92), Previous Close (\$409.43), and Open (\$414.41). Below the table, the Simple Price Viewer section with a 5-year time-range selector and three summary metrics is visible.

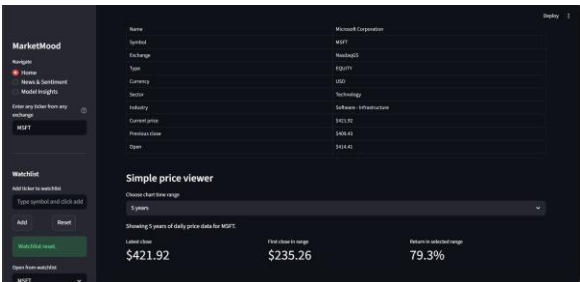


Fig. 6. Home Page – Instrument Details Table

Figure 7 displays the interactive Plotly closing-price chart for MSFT over the 5-year period (July 2021 to January 2026). The three period-return metrics show a latest close of \$421.92, a first close in range of \$235.26, and a total 5-year return of 79.3%. The chart clearly captures the post-pandemic rally (2021–2022), the 2022 correction, the 2023–2024 AI-driven bull run peaking near \$550, and the subsequent consolidation into 2026.

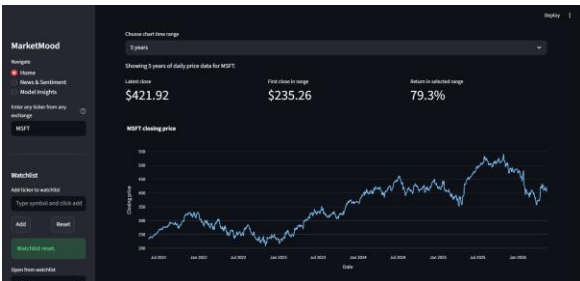


Fig. 7. Home Page – MSFT 5-Year Closing Price Chart

B. News Sentiment Analysis

Figure 8 shows the News & Sentiment page for MSFT with 10 of the most recent Finnhub headlines from 16–17 May 2026. VADER compound scores range from -0.3182 (Negative: “TTEC Stock Should Be on Your Radar...”) to $+0.25$ (Positive: “2 Nasdaq 100 Stocks with Exciting Potential...”). The majority of headlines (7 out of 10 visible) score exactly 0.0, assigned a Neutral label. This neutral-heavy distribution reflects the nature of informational financial headlines, which frequently employ flat, factual language that VADER’s lexicon scores as sentiment-neutral.

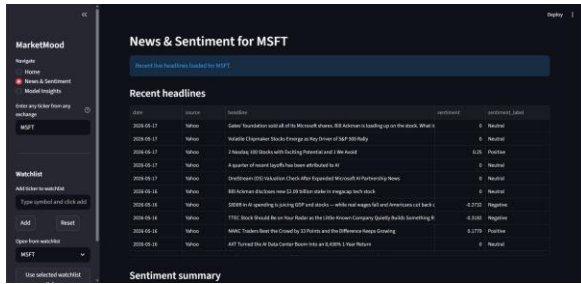


Fig. 8. News & Sentiment – VADER-Scored Headlines

Figure 9 shows the sentiment distribution histogram and the next-day prediction. The stacked bar chart reveals that the preponderance of MSFT headlines score in the neutral zone (compound score near 0), with two negative headlines at approximately -0.4 and three positive headlines at approximately $+0.2$ to $+0.5$. The resulting average compound score is mildly negative, driving the model’s prediction of a 55.6% probability of downward movement for MSFT the following trading session.

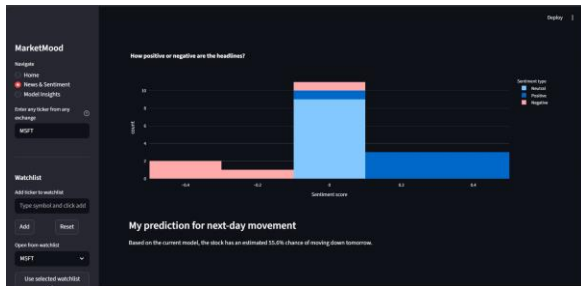


Fig. 9. Sentiment Histogram and Next-Day Prediction

C. Model Performance and Discussion

Figure 10 shows the Model Insights page for MSFT. The model comparison table reports 45.2% accuracy for both the DummyClassifier baseline and the Logistic Regression model on the chronological 25% test split of a six-month OHLCV window. The parity in accuracy between the two models is a significant and expected finding given the design of the sentiment feature: a single scalar average compound score, computed from the most recent batch of headlines, is broadcast identically to every row in the training matrix. This means that the sentiment feature

provides zero intra-dataset variance, rendering it informationally equivalent to a constant column and offering no discriminative signal to the logistic classifier beyond what is already captured by `return_1d` and `volume_change`.

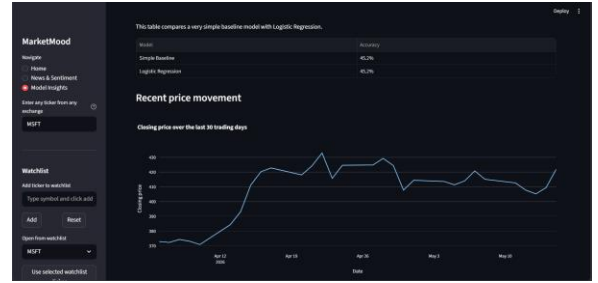


Fig. 10. Model Insights – Comparison and 30-Day Price Chart

This limitation is well-aligned with the literature: Loughran and McDonald [5] demonstrated that domain-general sentiment scores correlate poorly with financial outcomes, while Li et al. [7] showed that combining per-day, per-article sentiment time series with price features yields meaningful accuracy improvements. The current architecture intentionally simplifies the sentiment representation as a baseline to establish the deployment scaffold, with the explicit design intent of replacing the scalar average with per-day FinBERT embeddings as the primary future enhancement.

D. Comparison with Existing Systems

Table I benchmarks MarketMood against three representative systems from the literature.

TABLE I
Comparative Benchmarking of MarketMood

System	Data	NLP Method	ML	Deploy
AZFinText [4]	News articles	Noun-phrase SVM	SVM	Offline
Bollen et al. [1]	Twitter	OpinionFinder	SOFNN	Research
Li et al. [7]	News+price	Custom NLP	Multiple	Offline
MarketMood	YF+Finnhub	VADER live	Log. Reg.	Streamlit web

MarketMood is the only system in this comparison that integrates a live multi-source financial data pipeline, NLP-based sentiment scoring, supervised ML classification, and publicly accessible real-time web deployment with global ticker coverage in a single freely available tool.

VI. CONCLUSION & FUTURE WORK

This paper presented MarketMood, a sentiment-driven, next-day stock trend prediction platform that combines VADER headline sentiment scoring with engineered price features and Logistic Regression classification, deployed as a live, multi-page Streamlit web application. The system

demonstrates a fully operational, reproducible framework for sentiment-aware financial analysis that is globally accessible, requires no institutional data subscriptions, and supports any Yahoo Finance-compatible ticker including equities, ETFs, indices, and cryptocurrencies.

The experimental evaluation on MSFT confirms that the current three-feature logistic model achieves accuracy (45.2%) indistinguishable from the DummyClassifier baseline. This outcome is directly attributable to the use of a single scalar average as the sentiment feature, which provides zero row-level variance in the training matrix. This is an honest and reproducible baseline result that provides a well-defined starting point for future enhancement rather than overfitting to short historical windows.

The primary planned enhancement is the replacement of VADER with FinBERT, a BERT-based transformer pre-trained on financial communications, to produce per-article, per-day contextual sentiment embeddings that vary meaningfully across training rows and capture domain-specific financial language semantics. Additional directions include: gradient boosting classifiers (XGBoost, LightGBM) for non-linear decision boundaries; LSTM or Temporal Convolutional Networks trained on sequential price-sentiment time series; integration of social media streams from Twitter/X, Reddit WallStreetBets, and StockTwits to broaden the sentiment signal; real-time intraday data streaming for sub-daily prediction granularity; portfolio-level multi-ticker sentiment aggregation with correlation analysis; incorporation of earnings calendar events, options implied volatility, and macroeconomic indicator feeds as supplementary features; and cloud deployment on AWS or Streamlit Community Cloud for public, unauthenticated access.

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